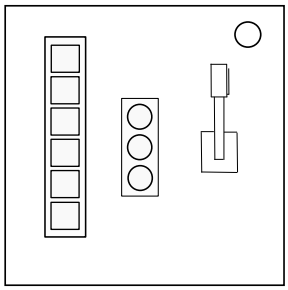
On the Subject of One-lever combination



It's only one lever, how hard could it be?

The module is composed of three LEDs, six black squares and a lever. Pulling the lever will make the squares gradually turn yellow; unpulling it will make them go back to black.

- Determine a five-number sequence using the five steps below.
- Flick the lever to start inputting the sequence.
- To input a number, pull the lever when the same number of yellow squares is present.
- If the lever is still pulled when all six squares are yellow, they will all turn black and the lever will be unpulled.
- Inputting the incorrect sequence will cause a strike. If the squares are all black again, the input will be reset without causing a strike.

Determining the number sequence

Note: the numbers go from 1 to 6, therefore if any number calculated using the steps below would be more than 6, then the number for that step is 6 instead. Similarly, if any number would be less than 1, then the number from that step is 1 instead.

Step 1:

- If the three LEDs are all on or all off, the number is 6.
- Otherwise, if only one LED is on, the number is 4.
- Otherwise, the number is 5.

Step 2:

- If the bottom LED is off and there are at least two batteries on the bomb, the number is the number from Step 1 minus the number of batteries.
- Otherwise, if the top LED is on and there are at least two ports on the bomb, the number is the number from Step 1 minus the number of ports.
- Otherwise, the number is the number from Step 1 minus 3.

Step 3:

- If the last digit on the serial number is odd and there is at least one lit indicator on the bomb, the number is the number from Step 2 plus the number of lit indicators.
- Otherwise, if the serial number contains a vowel and there are at least two unlit indicators on the bomb, the number is the number from Step 2 plus the number of unlit indicators.
- Otherwise, the number is the number from Step 2 plus 2.

Step 4:

- If the number from Step 3 is even, the number is the number from Step 3 minus 3.
- Otherwise, if the number from Step 2 is odd, the number is the number from Step 3 minus 1.
- Otherwise, the number is the number from Step 3 minus 2.

Step 5:

- If all LEDs are off, the number is 5.
- Otherwise, if the middle LED is the only one on or the only one off, the number is 6.
- Otherwise, the number is the sum of the numbers from Step 2 and Step 4.